

Feasibility Study Report: Multimodal Ball Action Anticipation

Using SoccerTrack v2

Thesis Feasibility Review | 20 August 2026

1. Purpose and Research Direction

This feasibility study evaluated whether a small undergraduate thesis team can construct and experimentally test a multimodal Ball Action Anticipation (BAA) system using public SoccerTrack v2 data and limited cloud compute. The proposed research question is: Does explicit player game-state information improve short-horizon ball action anticipation compared with a visual-only anticipation baseline? The intended input is a 30-second history of panoramic RGB video and synchronized player game state; the target is one or more ball actions occurring in the following 5 seconds.

The feasibility objective was deliberately narrower than full model development. We did not attempt to prove that fusion improves performance. Instead, we tested whether the required data exist, whether the modalities can be synchronized without leakage, whether valid anticipation samples can be generated, and whether visual and structured features can be produced under Colab-level resource constraints. A successful feasibility outcome means that model comparison is executable; improvement from fusion remains the research question.

2. Dataset and Feasibility Questions

SoccerTrack v2 was selected because it provides full-match panoramic video, Ball Action Spotting (BAS) annotations, and per-frame Game State Reconstruction (GSR). The Google Drive release was used because it was accessible in the working environment. An inventory of the release identified ten matches with video, GSR, BAS, and raw timing/calibration material: 117092, 117093, 118575, 118576, 118577, 118578, 128057, 128058, 132831, and 132877. The feasibility study used match 117093 as the representative deep-validation case.

The study was organized around five questions: (1) Are BAS targets and GSR states available with compatible identities and timing? (2) Can 30-second observation and 5-second future windows be generated without crossing period boundaries or duplicating targets? (3) Can a complete player-state tensor be materialized without future-frame leakage? (4) Does the released panoramic video align with GSR at the same match timestamps? (5) Can synchronized visual features be extracted on free Colab-scale compute?

3. Validation Procedure and Results

3.1 Dataset Inventory and Annotation Validation

The release inventory contained 22 video files, 20 GSR JSON files, and 11 BAS directory entries (including one non-data .DS_Store entry). All ten match identifiers listed above had the required BAS and GSR components. Missing MOT files in the Drive mirror did not block the chosen BAA route because player identity, team, role, image boxes, and pitch coordinates are already present in GSR.

For match 117093, the BAS file contained 2,252 actions involving 41 unique players. All twelve raw labels were present. The most frequent classes were PASS (951), DRIVE (889), and HIGH PASS (150), while HEADER (3) and GOAL (5) were extremely rare. This confirmed severe class imbalance. Therefore, class handling is treated as a later experimental decision rather than silently removing rare classes during feasibility analysis.

3.2 GSR Schema, Identity Coverage, and Period Timing

The GSR JSON uses a COCO-like structure with info, images, annotations, and categories. Player annotations include image_id, track_id, player_id, role, team, jersey number, image-space bounding boxes, and pitch coordinates. Match 117093 contains 67,650 first-half GSR frames and 70,475 second-half GSR frames at 25 FPS. The first-half sequence duration is 45:06 and the second-half duration is 46:59.

BAS actor identities were checked against GSR. Identity coverage was 100% for the BAS actor IDs observed in both halves (23/23 in the first half and 27/27 in the second half). This established that future-action actors can be related to the structured game state. Raw period

metadata further provided exact frameStart, frameEnd, matchTimeStart, and matchTimeEnd values. The interval counts exactly matched the GSR sequence lengths at 40 ms per frame, which allowed a frame-exact BAS-to-GSR mapping.

The verified mapping uses absolute BAS match time and period metadata: $\text{local_ms} = \text{BAS_position} - \text{period_matchTimeStart}$; $\text{local_frame} = \text{round}(\text{local_ms} / 40 \text{ ms})$; $\text{GSR image_id} = \text{first_image_id} + \text{local_frame}$. Boundary events tested with this rule produced zero-frame differences. The important finding is that second-half BAS times are absolute match times and must not be treated as restarting from zero.

3.3 Construction of the Derived Anticipation Benchmark

A 30-second observation window, 5-second anticipation horizon, and 5-second stride were used to construct a representative BAA benchmark. Windows were allowed only when the full observation and future interval remained inside the same half. For match 117093, this produced 1,092 valid reference times: 535 in the first half and 557 in the second half. Every observation had exactly 750 native GSR frames.

The future windows were naturally multi-label: 715 windows (65.48%) contained two or more actions, 144 (13.19%) contained one action, and 233 (21.34%) contained no action. This confirmed that the task cannot be simplified to only positive single-action windows. A provisional 10-class compatibility view excluding FREE KICK and GOAL was also generated for analysis, but this is not a final thesis class policy. Across the representative match, 2,197 compatible target assignments were produced; each covered action was assigned exactly once, with no duplicate assignments. The 33 eligible actions not covered were all located too early in a half to provide the required 30-second past context.

3.4 Structured Game-State Materialization and Leakage Test

A concrete sample, 117093_H1_0006, was materialized from 30.0 to 60.0 seconds, with the future target interval from 60.0 to 65.0 seconds. Its future target is a CROSS occurring 4.56 seconds after the prediction reference time. The observation contained 750 GSR frames and 22 tactical entities per frame (20 field players and 2 goalkeepers), with no missing observation frames, no missing x/y coordinates, and no duplicate player-frame records. All 22 players were present throughout the sample.

The canonical structured representation had `positions_m` (750, 22, 2), `positions_norm` (750, 22, 2), `presence_mask` (750, 22), role codes, and team codes. The latest represented GSR time was 59.96 seconds, the future interval began at 60.00 seconds, and the first target occurred at 64.56 seconds. Therefore, no future GSR frame leaked into the observation. A 5 FPS view retained 150-time steps and was used only as a lightweight feasibility sampling rate; 5 FPS is not yet claimed to be optimal.

3.5 RGB Video Validation and Diagnosis of the Frame-Count Mismatch

The first downloaded calibrated panorama was only 47 frames (1.88 s) and was rejected as an invalid full-half artifact. We then downloaded the canonical first-half panorama for match 117093. The file was approximately 3.35 GB, 4096 x 1080 pixels, 25 FPS, and 2706.002 seconds long. However, FFprobe reported 67,625 decoded video frames, whereas GSR contains 67,650 frames. This 25-frame difference initially prevented a naive frame-index alignment assumption.

A timestamp-level diagnostic resolved the discrepancy. The RGB stream begins at presentation timestamp (PTS) 1.000 s, ends at 2705.96 s, has exactly 67,625 decoded frames, and contains no large timestamp gaps. Frame intervals are consistently 0.04 s. By contrast, GSR represents the half from match time 0.00 s. Therefore, the video is internally continuous but begins one second later in its presentation timeline. This explains why equal decoded frame numbers cannot safely be used to synchronize the two modalities.

3.6 Direct RGB-GSR Synchronization Test

Synchronization was tested at match time 30.0 s. The RGB frame nearest PTS 30.0 s was decoded frame 725, whereas the GSR state for 30.0 s was local frame 750 (`image_id` 3000751). The GSR frame contained 22 tactical image-space bounding boxes. These boxes were overlaid on the RGB image at PTS 30.0 s and visually aligned with the corresponding players. As a control, the same 30.0-second GSR boxes were placed on RGB decoded frame 750, whose actual PTS is 31.0 s; this naive equal-index overlay showed visible displacement. The comparison provides direct empirical evidence that the correct synchronization method is timestamp-based, not raw frame-index equality.

3.7 Common Temporal Grid and Visual Feature Extraction

A common 5 FPS grid was then created from 30.0 to 59.8 s, yielding exactly 150 timestamps. RGB indices advanced by five decoded frames per step, and GSR frames advanced by five native frames per step. The maximum RGB timestamp matching error was approximately 1.07×10^{-13} seconds, which is floating-point noise. Thus, both modalities shared the same 150 observation times.

The 150 synchronized RGB frames were spatially reduced while preserving the panoramic aspect ratio and passed through a pretrained ResNet-18 used only as a feasibility probe. Feature extraction succeeded on CPU, producing a finite visual feature matrix of shape (150, 512). The synchronized structured branch at the same 5 FPS grid has 150 game-state time steps, e.g., positions of shape (150, 22, 2). This demonstrates that a real multimodal observation can be created under the available compute constraints. ResNet-18, 224-pixel height, and 5 FPS are not final architectural choices; they only establish execution feasibility.

4. Consolidated Feasibility Evidence

Feasibility item	Status	Evidence
Dataset availability	PASS	Ten matches contain the required video, BAS, GSR, and raw timing material.
BAS target usability	PASS	2,252 actions in match 117093; timing and actor IDs validated.
GSR state usability	PASS	Per-frame player identity, team, role, boxes, and pitch coordinates available.
BAS-GSR synchronization	PASS	Exact period timing rule and actor identity coverage validated.
30 s -> 5 s benchmark	PASS	1,092 valid windows; negatives and multi-action futures retained; no duplicate targets.
Leakage control	PASS	Observation ends at 59.96 s; future begins at 60.00 s in the tested sample.
RGB source usability	PASS	Full 4K, 25 FPS panoramic half-video successfully decoded.
RGB-GSR alignment	PASS	Timestamp-aligned player-box overlay correct; naive equal-index control is misaligned.
Common multimodal grid	PASS	150 exact shared timestamps over 30 s at provisional 5 FPS.
Visual feature extraction	PASS	Pretrained ResNet-18 probe produced finite (150, 512) features on CPU.
Fusion performance	NOT TESTED	This is the research experiment

5. Go/No-Go Decision

Decision: GO, with strict scope control. The major execution risks that could have invalidated the thesis route have been tested on a representative full-match case: data access, annotation schemas, actor identity coverage, period timing, anticipation-window construction, leakage control, structured-state materialization, real video decoding, RGB-GSR synchronization, and visual feature extraction. The multimodal pipeline can therefore be built under the available environment.

This GO decision does not claim that multimodal fusion will outperform a visual-only model. The remaining scientific uncertainty - whether explicit player game-state information improves short-horizon anticipation - is precisely the hypothesis to be tested. The representative match validation must also be extended to all ten matches during implementation; feasibility evidence from one match should not be reported as full-dataset empirical validation.

6. Experimental Design Implied by the Feasibility Study

The fair thesis comparison should use the same derived SoccerTrack benchmark, the same match-level train/validation/test logic, the same future targets, and the same visual preprocessing for all relevant models. A minimum comparison is: (A) visual-only, (B) game-state-only, and (C) visual + game-state fusion. If a relation-aware or graph model is introduced, a flat game-state control should also be included so that any benefit from message passing can be separated from the benefit of simply adding structured state.

Data splits must occur at the match level, never by randomly splitting overlapping windows, because 30-second windows with 5-second stride share substantial temporal content and would otherwise leak near-duplicate observations across partitions. Published visual BAA results from a different dataset/domain should be treated as methodological references rather than directly comparable scores. The visual baseline and multimodal model should be retrained or adapted on the same SoccerTrack-derived benchmark.

7. Remaining Risks and Scope Boundaries

- Only ten matches are available, so evaluation variance may be high. Match-level cross-validation or carefully justified match splits are important.
- Class imbalance is severe. The final rare-class policy, loss weighting, and whether to use the provisional 10-class compatibility view are not yet locked.
- The benchmark is derived from SoccerTrack BAS and is not an official standardized BAA benchmark. It must be described transparently as a derived anticipation benchmark.
- The final RGB backbone, resolution, sampling rate, and frozen-versus-fine-tuned strategy are implementation decisions. The ResNet-18/5 FPS probe should not be presented as the final architecture.
- A positive fusion gain is not guaranteed. A null or negative result remains scientifically interpretable if the baseline, controls, and evaluation are rigorous.
- Whole-dataset preprocessing, production caching, substitution edge cases, and full training were intentionally left outside feasibility and belong to implementation.

8. Conclusion

The feasibility study demonstrates that the proposed multimodal Ball Action Anticipation experiment is technically executable. On representative match 117093, BAS actions and GSR player state were synchronized, valid 30-second-to-5-second samples were constructed without leakage or target duplication, a complete structured-state sequence was materialized, the canonical panoramic video was decoded and timestamp-aligned to GSR, and a synchronized visual feature sequence was successfully extracted on CPU. The correct engineering rule discovered during testing is to align RGB and GSR by presentation/match timestamp rather than assuming identical raw frame indices. With these risks resolved, the next phase should be implementation of fair visual-only, game-state-only, and fusion baselines rather than additional feasibility engineering.

References

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- [2] SoccerTrack v2 Project Page. <https://atomscott.github.io/SoccerTrack-v2/>
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